Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

DRAFT — UE26CY151A · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry · Rotational spectroscopy

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Rotational spectroscopy
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Rotational spectroscopy

1 Rotational spectroscopy

Study focus: Rotational spectroscopy. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000046 · EV0000047

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module one on molecular spectroscopy and in today's class we are going to derive the expression for rotational energy levels of a diatomic molecule.

Key details

- We will study the selection rules and finally we will arrive at the rotational spectrum.
- We need a model in order to understand how the molecule is going to behave, and we can see here what we are going to use is a rigid rotor molecule.
- We have two atoms heteronuclear so the masses are different and they are held together by a rigid rod and that is having a center of gravity at the point C.
- which are held together by a rigid rod, meaning that whenever there is rotation, this rod will not change its form.
- Now we have masses M1 and M2 which are joined by a thin rod, so therefore R0 is going to be R1 plus R2.
- Therefore, we can write the balancing equation m1r1 is equal to m2r2.
- M2 R2 R1 plus M1 R1 R2.
- All we have done is replaced M1 R1 by M2 R2 and here M2 R2 by M1 R1.
- And what we get is I is equal to R1 R2 M1 plus M two.
- Because there is a rotation which is happening.
- Now since m1 r one is equal to m2 r2 we can also write this as m2 r naught minus r1 because we said that r naught is equal to r one plus r2.
- And why do we require this?
- Because we want to substitute R1 here.
- So therefore R2 is equal to M1 R0 by M1 plus M2.

- When we substitute for R1 and R2 in this expression for moment of inertia we get this expression i is equal to m1 m2 divided by m1 plus m2 r naught square which can be written as mu r naught square.

Example or application from the lecture

- So let us get to the expressions of rotational energy levels for a diatomic molecule, because that is the simplest molecule.
- So let us take the case of a diatomic molecule and heteronuclear diatomic molecule of course because homonuclear diatomic molecules will not be microwave active.
- Let us see how the derivation goes.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 05:00–09:59

I is equal to mu r naught square.

Key details

- And what is r naught?
- It is the distance between the two atoms or the two masses which we have considered here.
- We are still talking in terms of classical mechanics.
- Energy is half I omega squared.
- Omega is the angular velocity.
- This can be written as L squared by 2i.
- Because L is the angular momentum.
- because we are talking of a single molecule and here classical mechanics or classical laws are not applicable to molecules which are so small.
- What does this expression show us that the angular momentums are all quantized.
- They are not continuous.
- In the classical case we said L squared by 2i.
- L can have any value, but here L cannot have all the values.
- It can take up only those values which J has because.
- Here do j plus 1 by 2 pi h.
- So therefore the quantity j is known as the rotational quantum number and it can take up integral values 0, 1, 2, 3.
- So we will get different values of quantum the angular momentum based on the value of j which we substitute here and therefore we can see that angular momentum is quantized.
- How we can substitute for L in this expression for energy, and what we obtain is EJ.
- is equal to h squared by 8 pi squared i j g plus 1 joules.
- This energy is expressed in terms of joules and what is h here it is the Planck's constant 6.626 into 10 power minus 34 joules second.
- And I is the moment of inertia of the.
- Which energy are we talking about?
- The rotational energy levels inside the molecules is R context.

- Because they cannot have any value but those which are decided by the value of the rotational quantum number that is J.
- And we know that J can take up only values of 0, 1, 2, 3.
- If I want to convert the energy into wave number we need to divide by H C.
- That was H squared by 8 pi squared I, so it becomes H by 8 pi squared I C, J times J plus 1 centimeter inverse.
- Now this is a constant for a particular molecule because H pi and C are constant C is the speed of light and that is 3 into 10 to the power of 8 meters per second and i is the moment of inertia of the molecule which is again constant.
- Therefore this can be written in the form of a constant b.

Example or application from the lecture

- So therefore we will use the Schrödinger equation for the rigid rotor, and then we find when we solve the Schrödinger equation that angular momentum is now given by this expression root of j plus 1 by 2 pi h or we can say root of j plus 1 h cross which can also be written as h by 2 pi.
- Let us see what these energy levels look like.

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Core explanation · 10:00–14:59

1 centimeter inverse.

Key details

- And what is the value of B is H over A pi squared I C centimeter inverse and B is known as the rotational constant.
- Is known as the rotational constant and it's expressed in centimeter inverse, it will be h by a pi squared i c.
- And that's how we arrived at the expression for the rotational energy levels.
- This is the energy scale.
- And we have 0 is at g is equal to 0, we have energy as 0.
- When we substitute j is equal to 1, we get 2b.
- And that's what is shown here.
- And when you get go to j is equal to 2, we get 6b.
- We can see this is 0, 2b, 6b and so on.
- We can go to the next level that is 12b.
- If you want to find out for any energy level, all you have to know is the value of B.
- When we substitute the values for J 0, 1 to 3 we get the different energy levels.
- But just knowing the rotational level is not going to help us get the spectrum because what we want finally or what we get when we do this spectroscopy is the spectrum.
- We have to follow certain rules which are known as the selection rules.
- If we understand the selection rules, then we will be able to arrive at the spectrum.
- The selection rule for rigid rotor model.

- Is obtained after solving the Schrödinger equation.
- But there is another specific selection rule which is employed and which is applicable only for a rigid rotor heteronuclear diatomic molecule and that says delta J is equal to plus minus one.
- When it is plus one we refer to absorption spectrum.
- When it is minus one we refer to the emission spectrum.
- We know is equal to the rotational levels where are they?
- Bj plus 1 centimeter inverse.
- We have these energy levels inside the molecule.
- Just now we saw.
- Delta J is plus minus 1 means that.
- And we can have a transition zero to one, we can have a transition one to two, we can have a transition two to three, we can have a transition two to three to four, but we cannot have zero to two, two to four.
- All those transitions are forbidden.
- Only transitions allowed are between one level and the next level present in the molecule.
- Why am I interested in the energy gap?
- Because we know that the radiation will be absorbed based on the energy gap.
- Rotational levels, of course, we know where they are, but more important for us is to know what is the energy gap.
- What is the energy gap between J to J plus 1?
- So we substitute here J plus 1 first minus for so J.

Example or application from the lecture

- And these are rotational energy levels because we consider that the molecule is rotating.

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EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 15:00–19:59

wave number and that will be the expression becomes 2bj plus 1 centimeter inverse.

Key details

- It is going so J is equal to zero.
- That's what we mean.
- That is one.
- When we go to from one to two, it will be four b.
- When we go from two to three, it is going to be six b and so on so forth.
- We can calculate the energy.
- the energy gap between the energy levels and we see very clearly as we go to higher energy levels the energy gap is going on increasing.
- And if you have a transit and only delta j is equal to plus minus 1 is what is allowed.
- And here we can see plus 1 is what we are doing here.

- Zero then one to two, then it's going to be four b and then six b and eight b and ten b so on so forth.
- the energy gap itself is going on increasing.
- Energy will be absorbed.
- And what energy will be absorbed?
- That will be corresponding to the energy gap.
- So that means when radiation falls on this molecule, the first line will appear at 2b because that is the lowest energy gap, and remember this is in centimeter inverse this way.
- That means wave numbers energy is increasing.
- You can see from the energy level diagram here that this requires minimum energy.
- Energy and that will correspond to 2B.
- 0, 2, 1.
- Then we have 1, 2, 2, that would be 4b, and then we have 6b here, and then we have 8b, 10 P and 12B.
- The first line is appearing at 2b, but.
- The difference between these lines is also very clear.
- It is going to be 2b.
- of H C.
- So you will have the first line appearing at 2b.
- And the distance between any two lines in the spectrum is going to also be equal to 2b, and that remains constant.
- And from here we can calculate what is to be.
- Mu is the reduced mass which is fixed for any molecule which is mu is equal to we already know m1 m2 by m1 plus m2.
- And what is R0 here?
- What was our model?
- This is R0 and what is that?
- Will be for any molecule it is going to be the bond length.
- So that means.

Example or application from the lecture

- Let us see the spectrum.

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Core explanation · 20:00–24:59

Using I the microwave spectroscopy or rotational spectroscopy, we can determine the bond length of the molecule.

Key details

- You can see that the intensity of the lines is increasing and then.
- Intensity depends on the population of the energy levels and therefore we can see from this spectrum that it is not just the ground level which is the most populated.

- Why we are saying that because this refers to zero to one transition, but this is not.
- The most populated level at room temperature, even some of the higher levels are also populated.
- It is not just the lowest level or the ground state which is the most populated in the case of rotational levels, even some of the higher levels are also populated, as you can see from the intensity of the lines.
- So in the next class we are going to study another type of spectroscopy, which is the vibrational spectroscopy.

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EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Rotational spectroscopy transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rotational spectroscopy transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rotational spectroscopy transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9997; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Rotational spectroscopy transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

C Revision and retrieval practice